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Chapter 25
Poker Finance: Portfolio Theory and Backing Agreements

In addition to the Kelly criterion, many concepts from finance apply to the management of a poker bankroll and the selection of games, hi this chapter, we will examine two concepts of particular importance. The first is examining the volatility and expectation of a portfolio of poker activities. As a parallel to securities investing. certain types of diversification are helpful, while others are not. Second, we provide a framework for evaluating the expectation of a backing agreement between a player and a backer.

Portfolio Theory

In portfolio management, one critical statistic that is used to evaluate performance is the ***Sharpe ratio.*** The Sharpe ratio of a portfolio is the ratio of its expectation to its standard deviation. When investments are infinitely scalable (for most individual investors, stocks are), maximizing the Sharpe ratio of a portfolio leads to the best risk-adjusted return.

$$S = \frac{\mu}{\sigma} \qquad \textbf{(25.1)}$$

One of the main ways in which portfolio managers can maximize their Sharpe ratio is through the process of ***diversification;*** that is, by buying multiple securities that have a higher Sharpe ratio when bought together then separately. The following is a very short and simplified introduction. Those of you with experience in finance will recognize that we are leaving out many important facets of stock portfolio analysis; the point here is to explain in a rough way the principle without worrying about things that do not affect poker.

Consider two independent investments A and B. Both A and B have an expectation of $10 and a standard deviation of $100 (for the amount we have to invest). If we invest it all in A, we'll get $10 expectation with a $100 standard deviation. The same applies if we invest it in B. But if we invest half of it in A and half of it in B, then we'll still have an expectation of $10, but our standard deviation will be $100/$\sqrt{2}$, or about $70. By doing this, we have increased our Sharpe ratio from 0.1 to about 0.14.

Now let's consider a portfolio made up of A and a third investment C, C has an expectation of $5 and a standard deviation of $100. We can see immediately that C is a worse investment than A, since we can get $10 of expectation for $100 in risk by buying A. To maximize our Sharpe ratio, then, should we simply buy 100% of A? Perhaps surprisingly, the answer is no.

Let's evaluate a portfolio made up of 80% A and 20% C. Such a portfolio would have expectation of (0.8) ($10) + (0.2) ($5) or $9. It would have a standard deviation of

$\sqrt{((80)^2 + (20)^2)}$ = $82.46. So the Sharpe ratio for this portfolio would be 0.1091. This value is higher than the Sharpe ratio for a portfolio made up of only A, so it turns out to be better to mix the two investments, even though C is a worse investment in an absolute sense.

We can apply this reasoning to poker as well. Consider a situation outside of a pair of tournaments. One player (A) has an expectation of 1 buyin per tournament and a standard deviation of 9 buyins/tournament in his tournament. Another player has an expectation of 1buyin per tournament and a standard deviation of 12 buyins/tournament in a different tournament. The first player wants to maximize his Sharpe ratio. What is the optimal percentage to swap with the second player?

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Let 1- α be the fraction that these two players swap. Then A is investing in a portfolio P of *a* A and (1 - α) B. The win rate of that portfolio will still be 1 buyin per tournament, since all the components have that win rate.

We want to minimize the standard deviation of that portfolio, which we can also accomplish by minimizing the variance:

$\sigma P^2 = (9a)^2 + ((1 - a)12)^2)$

Taking the derivative with respect to *a* and setting to 0:

$0 = 162a - 288(1 - a)$
$0 = 450a - 288$
$a = 0.64$

A should swap 36% of himself in order to maximize his Sharpe ratio.

We can derive this result more generally. Say we have two investments with expectation w_1 and w_2, and standard deviations σ_1, and σ_2. We can normalize these bets to each other in terms of expectation by simply multiplying σ_1 by w_2/ w_1. We will call the scaled standard deviation s_1. If we do this, then both investments will have the same win rate w_2. We will mix these two investments in a proportion of a_1 of the first and a_2 of the second.

$a_1 = 1 – a_2$

The variance of the portfolio P will be:

$\sigma P^2 = (a_1^2)(s_1^2) + (a_2^2)(\sigma_2^2)$

We have omitted some steps here in deriving the variance. This equation is only valid for independent events, where there is no correlation. When analyzing securities, correlation is very important. Large-field tournament play, even in the same tournament, for players playing to maximize their own expectation is only correlated a very tiny bit, so we omit that for the purposes of this discussion.

Since $a_2 = 1 – a_1$ we have:

$\sigma_P^2 = (a_1^2)(s_1^2) + (1 – a_1)^2(\sigma_2^2)$

Taking the derivative with respect to al and setting to 0:

$0 = 2a_1s_1^2 - 2(1 – a_1)^2(\sigma_2^2)$
$2\sigma_2^2 = 2a_1s_1^2 + 2a_1\sigma_2^2$

$$a_1 = \frac{\sigma_2^2}{\sigma_2^2+s_1^2} \qquad \textbf{(25.2)}$$

We can also easily find a_2:

$$a_2 = \frac{s_1^2}{\sigma_2^2+s_1^2} \qquad \textbf{(25.3)}$$

In order to maximize the Sharpe ratio of a portfolio, these are the percentages of each investment

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of which the portfolio should be composed. In finance, this type of analysis is extended to an arbitrary number of different investments and optimal portfolios can be constructed. In the same way, poker players can reduce their volatility while preserving their Sharpe ratio by swapping action, either in tournaments or in ring games. This can even be true if one of the players has higher expectation: that player can still benefit Sharpe-wise by swapping with the other player.

Those familiar with stock analysis might be tempted to think that other types of diversification might help to improve portfolio performance: for example, playing different types of games (sit-and-go tournaments, multi-table tournaments, limit ring games, etc.). We believe that playing many games is valuable for a number of reasons, including the flexibility to take advantage of temporary situations (such as when a very weak and very rich player is playing his favorite game for high stakes), and developing a more general view of poker. But portfolio management is not one of the reasons. The reason that this type of diversification doesn't lower volatility is that it isn't really diversification at all.

When we own a stock, we own it 100% of the time. When we own two stocks, we own both of them 100% of the time, but only 50% as much. However, when we play two different types of poker, it is analogous to a situation where we buy a stock, hold it for half the time, and then sell it and buy another stock. As a result, playing different types of games doesn't reduce variance directly. However, one related concept can lower variance while retaining the same win rate: that is online multi-table play. If a player can concentrate adequately on multiple games, he can lower his variance by, for example, playing two games of $10-$20 instead of one game of $20-40. Of course, he might be able to play two games of $20-$40, so he might be sacrificing equity. But by playing two games of smaller stakes, we can achieve similar expectation with lower variance, thus improving our Sharpe ratio.

Backing Agreements

The last topic for Part IV, a discussion of the mathematics of backing agreements, is a topic that is of some interest for those members of the poker community who are interested in backing players. Player backing agreements have existed as long as there have been players who lack sufficient bankroll to play the stakes they wish to. The usual form of these agreements is this: the player receives some sum of money (or a smaller amount, usually with a promise to advance more if needed up to a certain point) from the backer. The player plays some agreed-upon game or games, and then at certain points, the agreement settles and some prespecified division of the profits takes place. If there are losses, the backer usually absorbs them all, but the player must make up any previous losses before cashing out any profit. There may also be a salary component, in which case the player's share of profits is often reduced dramatically.

As our starting point, we take as a simple backing agreement the following:

- The player has a known win rate in a particular game of $40/hr and standard deviation of $600/hr but no bankroll.
- The backer and player agree that the duration of the agreement will be 100 hours of play and the backer will fund all play. At the end of this, the profits will be divided evenly, or if there are losses, the backer will take them all, and the agreement will be terminated (although both parties may agree to enter into another agreement).
- The player agrees to play to maximize play EV instead of his own personal EV

Over the past two chapters we borrowed heavily from the finance world. In evaluating the value of backing agreements, we do so again. This backing agreement, then, takes this form. "We will randomly select some value from a prespecified probability distribution. If the value selected is below zero, then the value to the player is zero. If the value selected is above zero, then the value

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to the player is $^1/_2$ the value selected." This is equivalent to an ***option.*** In stock trading, an option is a contract that gives its holder the right to either buy a stock at a particular price, or sell the stock at a particular' price before a certain time. To the player, this backing agreement is essentially a "call" option (where the player may buy half the result) at a strike price of zero.

We can use this to find the value of the backing agreement to the player. The distribution of outcomes for 100 hours of play is normally distributed around a mean of $4,000, with a standard error of $6,000. The value to the player of this agreement depends on the value of the player's result if he wins. To find this value, we can integrate the product of the player's result (x) and the normal distribution density function from zero to ∞ to get the value of the player's win when he wins.

$$\frac{1}{\sigma\sqrt{2\pi}}\int_0^\infty xe^{\frac{-(x-w)^2}{2\sigma^2}}dx \qquad \textbf{(25.4)}$$

We can make the substitution x = x + *w* - *w* to obtain:

$$\frac{1}{\sigma\sqrt{2\pi}}\int_0^\infty [(x-w)e^{\frac{-(x-w)^2}{2\sigma^2}} + we^{\frac{-(x-w)^2}{2\sigma^2}}dx]$$

We can split this into two integrals. The second integral can be simplified by factoring out the *w* and the remaining expression is just the cumulative normal distribution:

$$\frac{1}{\sigma\sqrt{2\pi}}\int_0^\infty (x-w)e^{\frac{-(x-w)^2}{2\sigma^2}}dx + w\Phi\left(\frac{w}{\sigma}\right) \qquad \textbf{(25.5)}$$

To find the antiderivative of the expression $(x-w)e^{\frac{-(x-w)^2}{2\sigma^2}}$, we make the substitution $u = \frac{-(x-w)^2}{2}$, du = - *(x - w)*. This leaves us with:

$$\int -e^{\frac{u}{\sigma^2}}du = -\sigma^2 e^{\frac{-(x-w)^2}{2\sigma^2}} + C$$

We can now simplify Equation 25.5:

$$\frac{1}{\sigma\sqrt{2\pi}}\int_0^\infty (x-w)e^{\frac{-(x-w)^2}{2\sigma^2}}dx + w\Phi\left(\frac{w}{\sigma}\right)$$

$$= \frac{1}{\sigma\sqrt{2\pi}} - \sigma^2 e^{\frac{-(x-w)^2}{2\sigma^2}}]_0^\infty$$

$$< wins >= \frac{\sigma}{\sqrt{2\pi}}e^{\frac{-w^2}{2\sigma^2}} + w\Phi(\frac{w}{\sigma}) \qquad \textbf{(25.6)}$$

Equation 25.6 is the value of the entire option; that is, it is the value of the agreement if the player gets 100% of the win and takes none of the loss. To convert this value to the value of the backing agreement, we simply multiply by the player's share.

For our particular player, we can solve for the total value of the wins:

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w = 4000
σ = 6000
z = $(^{4000}/_{6000}) = {}^{2}/_{3}$

$(6000/\sqrt{2\pi})(e^{-0.22222}) + (4000)\ (\Phi(^{2}/_{3})) = \$4{,}906.72$

Since we know that the total expectation for the player must be $4,000, we can find that the value of the losses is -$906.72.

For our example backing agreement, the player's expectation, then, is half of the win and none of the loss, or $2,453.36. while the backer's share is all of the loss plus half of the win, or $1,547.64. Thus the player gets about approximately 61% of the expectation of the backing agreement.

This methodology can be used to cover any backing agreement of this type. We can extend equation 25.6 by creating a threshold value a such that the player's share of the profits begins with wins above a by simply integrating from a to ∞ Instead of from zero.

Then we have:

$$< wins >= \frac{\sigma}{\sqrt{2\pi}} e^{\frac{-(w-a)^2}{2\sigma^2}} + w\Phi(\frac{w-a}{\sigma}) \qquad \textbf{(25.7)}$$

If we used the same backing agreement as before but the player's share only began at wins of $1,000 or more, for example, then we would have:

$(6000/\sqrt{2\pi})(e^{-.125}) + (4000)\ (\Phi(^{1}/_{2})) = \$4{,}878.24$

Obviously this value is smaller, since wins from $0 to $1,000 are excluded from its value. Looking at various values for the threshold, we can see a potential negotiating tactic for the player. Consider a player who proposes as follows:

"Instead of getting half the profits when I make more than zero, I think that I should only be cut in if I exceed the expectation that we've agreed upon. However, since I'm giving up some money for those times that I win less than my expectation, I should get a larger piece of the profits above expectation."

This line of reasoning is sound. But how much higher a percentage of the profits should this player receive? We invite the reader to test his intuition here before continuing to see the answer.

The value of the option with a threshold of $4,000 is:

$(6000/\sqrt{2\pi})(e^{0}) + (4000)\ (\Phi(0)) = 4{,}393.65$

If the player was to receive 50% of the value of the wins at threshold $0, he would receive $2,453.36. To reach parity with that value, he "deserves" 55.8% of the win above $4,000.

Another technique that can be applied here is creating a tiered structure, where the player receives different percentages of amounts won, usually increasing as the player's win grows. Consider the following backing agreement:

- The player has a known win rate in a particular game of $20/hr and standard deviation of

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$240/hr.

- The backer and player agree that the duration of the agreement will be 100 hours of play and the backer will fund all play. At the end of this time, any losses will be absorbed by the backer, If there are profits, they will be divided according to the following schedule:

Profit Tier	**Backer Share**	**Player Share**
0 - $1000	75%	25%
$1000 - $2000	60%	40%
$2000 - $3000	50%	50%
$3000 - $4000	40%	60%
$4000 - $6000	30%	70%
$6000+	25%	75%

In the above table, the backer/player shares are what the player earns on that amount of profit. For example, if the player's result is $1,500, he earns 25% on the first $1,000 and 40% on the remainder, or $450.

We can calculate the value of such an agreement using Equation 25.7,

w = $2,000
σ = $2,400

The value of the total option for the first tier ($0-$1,000) is simply the value of the total option with threshold 0 minus the value of the total option with threshold 1000:

$(2400/\sqrt{2\pi})(e^{-.34722}) + (2000)(Z(^5/_6)) = \$2,271.93$
$(2400/\sqrt{2\pi})(e^{-.0868l}) + (2000)\ (Z(^5/_{12})) = \$2,200.93$

The difference is $71. This is the value of the total option for the first tier. We can do the same sorts of calculations to find the values of each tier:

Profit Tier	**Backer Share**	**Player Share**	**Tier Value**	**Value to Player**
$0 - $1000	75%	25%	$71.00	S17.75
$1000 - $2000	60%	40%	$243.47	$97.39
$2000 - $3000	50%	50%	$402.69	$201.34
$3000 - $4000	40%	60%	$473.53	$284.12
S4000 - $6000	30%	70%	$746.02	$522.84
$60000+	25%	75%	$334.33	$250.74
Total			**$2,271.93**	**$1,374.19**

This is roughly equivalent to a 60.4% share in the profits above zero.

This last example shows an alternative method for structuring backing agreements that might be slightly more effective at solving one of the two big issues facing backing agreements hi general. These two issues arc at the core of deciding how to structure a backing agreement. The first issue is the problem of ***aligning the interests of the player with the interests of the backer.***

One common problem that occurs quite frequently in backing agreements of a fixed term is that when the option expiry date (the end of the backing agreement) nears, if the player is at least

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moderately far away from the backing agreement threshold, the player has a direct incentive to take on a lot of variance. To take an extreme example, assume this is the last hand before the backing agreement expires. The player is $2,000 away from the threshold value where he begins to get shares of the profits. The player is offered an opportunity to put $3,000 in the pot with QQ against AA preflop (assume the AA player accidentally exposed his hand or the like.) This would be a terrible play for the backer and under normal circumstances, and the player as well. However, the player's EV from gambling here is actually about 18% of his split of $1,000, while his EV from folding is zero. Here the player's EV function is dramatically different and in this extreme case effectively opposed to the backer's.

One way of addressing this problem that has been used in some circumstances involves paying some sort of salary to the player that he receives win or lose, while reducing the player's share of the outcomes. However, this has the added effect of making it less valuable for the player to play well and try to win. This problem is familiar to managers in many sales industries: some automobile dealerships have experimented with commissioned employees and salaried salespeople and so on.

The second issue is that of ***fairly compensating both parties.*** Both sides want to make money, and presumably the backing agreement should benefit both parties. But what elements should be included hi this "fair" division of profit? Some principles that merit consideration are:

- A risk premium for the backer.
  The backer is assuming all the risk in these agreements and therefore ought to receive a premium for that risk-taking.
- The player may not always need backing.
  Often players who are being backed could instead play at some lower limit for some reduced rate. The player may feel that his gain from the backing agreement should reflect this. Viewed another way, this is like payment for the player's time, which could be spent in some other way. Since the backer is effectively making money passively on the deal, the player may demand a larger share of the profits.
- Uncertainty about the player's win rate.
  Often in creating backing agreements, some implicit or explicit win rate and standard deviation will be used. However, if the player's win rate is actually ***less*** than the parties believe, then the player's share of the overall value of the option increases. For example, assume 100 hours of play with w = $20 and σ = $200, compared to 100 hours of player with w = $15 and σ = $200. The total value of the player's play is now three-fourths as much. However, the value of the 0-threshold option is 81.3% as much with w = $15 as with w = $20. The player retains a higher percentage of his share of the win than the backer if his actual win rate turns out to be lower than the parties believe *a priori*.

None of these considerations include other items such as the risk of credit default or fraud, where the player absconds with the backer's money or fails to accurately report results. Of course, backing is a high-risk activity; however, the returns from a strong, stable backing agreement can far outstrip returns from other types of investments. Another problem with backing is that players who win often don't need to be backed for long; by contrast, players who are break-even or win slightly often require backing for longer periods of time in order to be profitable.

Key Concepts

- A poker bankroll shares many characteristics with a portfolio, and so elements of portfolio theory can be brought to bear on it.
- Sharpe's ratio provides a metric for risk-adjusted return. The higher the ratio, the higher the return relative to risk of the investment.

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- Players can benefit utility-wise from swapping with each other. This is even true if one player is stronger than the other but the swap takes place at par.
- Backing agreements share many characteristics with options, and can be valued similarly in deciding on a fair deal.

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Part V: Other Topics

There are more things in heaven and earth, Horatio, Than are dreamt of in your philosophy.

William Shakespeare, *Hamlet, I.v*

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